

6	B	Motorcycle travels in the same direction during the whole duration. Impulse or the change in momentum is the area under the force-time graph. $\Delta p = \int F dt$ $(400)(v - 4.5) = \frac{1}{2}(1.0)(400) - \frac{1}{2}(2.0)(800)$ $v - 4.5 = -1.5$ $v = 4.5 - 1.5 = 3.0 \text{ m s}^{-1}$
7	D	Since cube is floating, there is vertical equilibrium